COGNITIVE MECHANISMS OF CONSTITUTIONAL PERSISTENCE: A BAYESIAN FRAMEWORK BRIDGING INDIVIDUAL LEARNING AND INSTITUTIONAL LOCK-IN

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ABSTRACT

Why do constitutional provisions resist reform despite persistent dysfunction? Argentina's Article 14bis labor protections have survived 93 years and 23 reform attempts across seven ideologically diverse governments, generating 8.2% average unemployment and 47% informality—yet Brazil, Chile, and Spain successfully reformed comparable provisions. Existing theories fail: veto player models predict Argentina should reform more easily than Brazil; path dependence describes but doesn't explain; interest group theories face the union density paradox.

This article proposes **Heteronomous Bayesian Updating (HBU)**: humans learn legal norms primarily from observing authoritative reactions (parental disapproval, judicial precedent, professional sanctions) rather than behavioral outcomes. During legal education, students observe convergent high-authority reactions forming strong priors. These priors generate **extended phenotype structures**—187 specialized courts, 3,847 union legal departments, universal curricula—that defend founding doctrines through distributed active inference. When reforms create prediction errors, the HBU framework predicts actors systematically prefer blocking mechanisms over belief revision due to asymmetric coordination costs.

Quantitative validation across three independent datasets confirms predictions. Analysis of Argentina versus successful reformers shows correlation between extended phenotype density and reform resistance. Examination of 60 international law conflicts (2000-2025) reveals crisis conditions trigger 80% institutional blocking versus 0% under control (χ^2=36.736, $p<0.0001$). Constitutional memory extinction analysis demonstrates systematic precedent erasure despite formal validity.

This framework bridges cognitive science and institutional legal theory through falsifiable predictions. While requiring direct cognitive measurement, converging evidence across legal domains suggests HBU captures genuine regularities in institutional cognition.

(GitHub Repository: https://github.com/adrianlerer/Argentina-Labor-Regime-Analysis-2025)

Keywords: Bayesian inference, constitutional lock-in, active inference, extended phenotype, institutional persistence, legal evolution, predictive coding, reform resistance

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ARTIFICIAL INTELLIGENCE DISCLOSURE

This research was conducted entirely by the author without use of AI-generated content for theoretical development, data analysis, or substantive writing. Computational tools (Python 3.9, R 4.2) were used exclusively for statistical calculations, data visualization, and formatting verification. ChatGPT-4 and Claude 3.5 were consulted during final editing for grammar checking, sentence clarity, and detection of typographical errors only. All theoretical frameworks, empirical analyses, data collection, interpretations, and conclusions represent original work by the author. The author takes full responsibility for the accuracy and integrity of all content herein.

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I. INTRODUCTION

A. The Puzzle of Irreversible Dysfunction

On December 29, 2017, Argentine President Mauricio Macri signed Decree 27/2018, initiating what his administration characterized as the country's most comprehensive labor reform since the 1957 constitutional convention. The decree attempted to modify collective bargaining procedures and dismissal requirements embedded in Article 14bis of the National Constitution—a provision establishing workers' rights to "protection against arbitrary dismissal," "stable employment," and collective organization that Argentine courts had interpreted over six decades as mandating near-absolute job security with judicial pre-approval for dismissals. Within ninety-six hours of the decree's publication, labor tribunals in five provinces issued preliminary injunctions blocking implementation. Within three months, facing coordinated judicial resistance, extensive academic criticism, and unified opposition from the nation's 185 bar associations, the Macri administration quietly withdrew the reform. The constitutional provision, functionally unchanged since 1957, survived its latest challenge.

This outcome was neither anomalous nor surprising. Since Article 14bis entered constitutional force in May 1957, Argentina has attempted major labor law reform under seven distinct governments representing dramatically different ideological orientations: the developmentalist Frondizi administration (1958-1962), the bureaucratic-authoritarian Onganía dictatorship (1966-1970), the military junta under Videla (1976-1983), the neoliberal Menem government with three separate attempts (1991, 1995, 2000), the center-right De la Rúa coalition (2001), the market-oriented Macri presidency (2015-2019),

and most recently the libertarian Milei administration (2023-present). All seven attempts failed. The constitutional provision remains substantively unmodified for ninety-three years, despite Argentina experiencing labor costs 2.3 times the Latin American regional average, unemployment persistently above 7% since 1974, informal employment encompassing 47% of the workforce as of 2024, and GDP per capita growth rates substantially below regional comparators over the same period.

The persistence becomes more puzzling when examined comparatively. Brazil, Chile, and Spain—all possessing similar constitutional labor provisions adopted during roughly the same historical period and all facing comparable economic pressures—successfully reformed their labor regimes during the same timeframe that Argentina repeatedly failed. Brazil reformed its Consolidação das Leis do Trabalho (CLT) in 2017 after seventy-three years of stability, substantially modifying dismissal procedures, collective bargaining structures, and working time regulations. Chile reformed Articles 306 through 381 of its Labor Code six times between 1990 and 2023, each reform materially altering employment relationships. Spain reformed the constitutional framework governing Articles 35-38 through legislative interpretation and statutory modification five times between 1994 and 2022, including major restructurings during the 2008-2012 financial crisis. Yet Argentina, facing arguably more severe labor market dysfunction than any of these comparators, achieved zero successful reforms across the same period.

Research Question: What cognitive mechanisms explain why constitutional provisions resist reform despite persistent dysfunction, and why does this resistance vary systematically across jurisdictions with similar formal institutions?

Hypothesis: This article proposes that constitutional lock-in operates through Heteronomous Bayesian Updating (HBU), where legal professionals acquire strong priors through observation of authoritative reactions rather than outcome frequencies, then defend these priors via extended phenotype structures that minimize prediction errors through active inference (blocking reforms) rather than perceptual inference (updating beliefs). The framework predicts that jurisdictions with higher extended phenotype density should exhibit greater reform resistance independent of formal veto player configurations.

Constitutional scholars typically explain persistent institutional resistance through three well-established theoretical frameworks. Veto player theory (Tsebelis 2002) identifies the number and ideological distance of actors capable of blocking change, predicting that reform becomes increasingly difficult as veto players multiply and polarize. Yet this framework struggles with the Argentina-Brazil comparison: both countries require congressional supermajorities for constitutional amendment (Argentina requires two-thirds

approval; Brazil requires three-fifths approval in two separate votes plus presidential assent), yet Brazil succeeded while Argentina failed repeatedly. Path dependence theory (North 1990; Pierson 2000) describes how initial institutional choices constrain subsequent options through increasing returns, network effects, and coordination expectations. This approach effectively describes the phenomenon—institutions persist despite dysfunction—but provides no mechanism explaining how past choices exercise such powerful constraints on present actors who clearly perceive and suffer from the dysfunctional consequences. Interest group capture theories (Olson 1965) identify organized labor as the blocking force, but face an empirical anomaly: Argentine union density (25% of formal sector workers) is substantially lower than in Sweden (67%), Denmark (67%), or Finland (64%), all of which reformed labor laws multiple times since 1990. If union power alone explained resistance, high-density countries should exhibit greater reform difficulty than Argentina, yet the opposite pattern appears.

B. Theoretical Contribution

This article proposes a mechanistic framework bridging cognitive science and institutional legal theory. The core claim is that constitutional persistence operates through identifiable cognitive mechanisms scaling from individual learning to collective institutional behavior. Rather than treating legal institutions as sui generis phenomena governed by political or normative principles, this framework models them as extended phenotype structures (Dawkins 1982) constructed by humans executing learned behavioral programs, defended through distributed Bayesian inference processes minimizing free energy (Friston 2010).

Three conceptual innovations structure the analysis. First, I introduce Heteronomous Bayesian Updating (HBU): a learning mechanism whereby humans acquire normative beliefs primarily from observing authoritative reactions rather than behavioral outcome frequencies. Unlike standard Bayesian updating from direct experience, HBU operates through social observation of others' reactions to norm violations. A child learns "stealing is wrong" not from personal experience with theft outcomes but from observing parental disapproval; a law student learns constitutional interpretation not from testing doctrines empirically but from observing which interpretations professors reward or sanction. This mechanism generates strong priors resistant to contradictory outcome evidence because the priors were never grounded in outcome observation.

Second, I apply active inference theory (Friston 2013) to institutional cognition. Active inference distinguishes two strategies for minimizing prediction error: perceptual inference (updating beliefs to match observations) and active inference (acting to make observations match beliefs). Individual organisms and institutional collectives face identical choice: when

reality contradicts expectations, change expectations or change reality? The framework predicts systematic preference for active inference when perceptual inference faces high coordination costs. For constitutional provisions defended by 187 courts, 65 law schools, and 50,000+ legal professionals, simultaneous belief revision across this distributed cognitive system requires prohibitive coordination; blocking reforms through existing procedures requires minimal coordination. This cost asymmetry generates systematic resistance.

Third, I treat legal institutions as extended phenotypes—material artifacts constructed by replicating information patterns to serve those patterns' persistence rather than host organisms' welfare. Beaver dams are extended phenotypes of dam-building genes; they persist because they facilitate gene replication, not because they benefit individual beavers. Constitutional provisions are extended phenotypes of legal doctrinal packages (memeplexes); they persist because they facilitate memetic replication through professional training systems, not because they benefit constituent populations. Article 14bis survives ninety-three years not despite generating dysfunction but because it successfully built replication machinery (specialized courts, mandatory curricula, professional sanctions) that perpetuates founding doctrines across generations of legal professionals.

Methodological Note on Conceptual Integration

This article applies concepts from evolutionary biology and computational neuroscience to legal institutions. Three clarifications are essential to distinguish literal application from loose analogy:

First, extended phenotype (Dawkins 1982) is used literally, not metaphorically. Legal institutions are material artifacts constructed by humans executing learned behavioral programs, precisely as beaver dams are material artifacts constructed by beavers executing genetically-coded programs. The difference lies in transmission mechanism (cultural versus genetic) rather than phenotypic extension mechanism. Both represent information patterns constructing environmental structures that facilitate their own replication.

Second, free energy minimization (Friston 2010) describes computational architecture implemented in neural tissue. When applied to institutions, it describes distributed computation across multiple brains executing coordinated inference rather than metaphorical "institutional goals." The mechanism scales from individual to collective cognition because institutions are literally networks of individual cognitive systems running coordinated inference processes.

Third, Bayesian updating is not merely analogical but describes actual probabilistic inference performed by legal reasoners adjusting belief strengths based on observed evidence (precedents, doctrinal reactions, professional sanctions). Legal reasoning implements approximate Bayesian inference whether or not practitioners explicitly conceptualize it as such.

The framework thus bridges domains by identifying common computational principles implemented across different substrates (individual brains, institutional networks) rather than by drawing loose metaphorical parallels.

C. Contribution to Literature

This article makes three contributions. Theoretically, it provides a mechanistic account of constitutional lock-in grounded in cognitive architecture, explaining how past institutional choices constrain present actors (addressing the mechanism gap in North 1990; Pierson 2000). Previous path dependence scholarship describes institutional persistence without specifying cognitive processes implementing persistence. HBU offers that specification: past choices constrain present actors because those past choices installed learned behavioral programs executed automatically by current legal professionals, defended by extended phenotype structures those professionals constructed.

Empirically, the article offers quantitative validation across three independent legal domains. Analysis of constitutional reform attempts (Argentina versus Brazil, Chile, Spain) tests whether extended phenotype density predicts reform resistance independent of formal veto player configurations. Examination of 60 international law conflicts (2000-2025) tests whether crisis conditions trigger predicted shift toward active inference blocking. Investigation of 19th Amendment precedent extinction tests whether formally valid provisions can suffer functional extinction through distributed inference processes. Converging evidence across these domains supports the hypothesis that HBU mechanisms operate across legal contexts.

Methodologically, the article demonstrates how evolutionary biology and computational neuroscience can generate falsifiable predictions about institutional behavior. Rather than treating constitutional law as interpretive or normative enterprise, this framework models it as empirical science of institutional cognition. The approach offers a template for future interdisciplinary constitutional scholarship grounding institutional analysis in cognitive mechanisms.

D. Roadmap

The analysis proceeds in seven parts. Section II develops the theoretical framework: Heteronomous Bayesian Updating, active inference institutional cognition, and extended phenotype formation. Section III presents quantitative analysis of Argentina's Article 14bis persistence, constructing the Constitutional Lock-in Index (CLI). Section IV examines comparative cases (Brazil, Chile, Spain) testing whether CLI predicts reform success independent of veto player configurations. Section V analyzes 60 international law conflicts testing crisis-triggered active inference predictions. Section VI investigates 19th Amendment precedent extinction as legal memory erasure. Section VII discusses implications, limitations, and alternative explanations. Section VIII concludes with theoretical synthesis and research agenda.

II. THEORETICAL FRAMEWORK: HETERONOMOUS BAYESIAN UPDATING AND INSTITUTIONAL COGNITION

A. Bayesian Brains and Normative Learning

Contemporary cognitive science models human cognition as hierarchical Bayesian inference (Clark 2013, 2016; Hohwy 2013). Brains construct internal models of environmental structure, generate predictions about sensory inputs, and update models when predictions fail. This predictive processing architecture operates across perception, action, and abstract reasoning. Learning occurs through Bayesian belief revision: observing evidence updates probability distributions over hypotheses according to Bayes' theorem.

Standard Bayesian learning assumes agents update beliefs from direct experience with environmental contingencies. A rat learns food location by exploring environments and updating beliefs based on reward observations. An economist learns inflation dynamics by observing price sequences and updating macroeconomic models. This direct experiential updating characterizes much animal learning and some human learning in domains permitting individual experimentation.

However, human normative learning—acquiring beliefs about appropriate behavior—operATES differently. Humans rarely learn social norms through personal experimentation with norm violations. A child does not learn "stealing is wrong" by repeatedly stealing and observing outcomes; rather, the child observes parental reactions to theft (disapproval, punishment) and infers normative content from those reactions. A law student does not learn constitutional interpretation by testing competing doctrines in actual constitutional litigation; rather, the student observes which interpretations professors reward during classroom discussions and which interpretations generate sanctions (low grades, professional exclusion).

I term this mechanism Heteronomous Bayesian Updating: learning normative content from observing others' reactions rather than experiencing outcomes directly. The term "heteronomous" (governed by external authority) contrasts with "autonomous" learning from self-generated experience. HBU involves three components:

1. Observation of authoritative reactions: The learner observes how high-status actors (parents, teachers, judges, senior professionals) respond to norm violations or doctrinal positions.
2. Inference of underlying norms: The learner infers normative content from reaction patterns. Convergent negative reactions to behavior X generate strong prior that "X is impermissible"; convergent positive reactions to doctrine Y generate strong prior that "Y is correct."
3. Prior formation resistant to outcome evidence: Because priors were formed from reaction observation rather than outcome observation, contradictory outcome evidence (dysfunction, inefficiency) generates weak updating pressure. The prior "Article 14bis mandates job security" formed from observing unanimous judicial/academic endorsement remains strong even when economic outcomes demonstrate labor market dysfunction.

This framework builds on Cristina Bicchieri's social norms theory (Bicchieri 2006, 2016), which distinguishes personal normative beliefs from expectations about others' beliefs and sanctions. HBU specifies the learning mechanism generating coordinated normative expectations: observing authoritative reactions creates common knowledge of norm content and common knowledge that violations trigger sanctions.

B. Active Inference: Two Strategies for Minimizing Prediction Error

Karl Friston's free energy principle (Friston 2010, 2013) formalizes biological systems as minimizing surprise (prediction error) through two mechanisms:

1. Perceptual inference: Update internal models to better predict observations. When sensory input contradicts predictions, revise beliefs.
2. Active inference: Act on environment to generate observations matching predictions. When sensory input contradicts predictions, change the world.

This dual mechanism appears across biological organization. A hungry organism faces prediction error (predicted: food, observed: no food). It can resolve this through perceptual inference (update belief: "food not available here") or active inference (move to location where food is available). Both strategies minimize free energy—the divergence between predicted and observed states.

For simple organisms with limited cognitive capacity, perceptual inference dominates. But organisms capable of environmental manipulation increasingly deploy active inference. Humans minimize surprise extensively through active inference: building shelters (making temperature observations match thermal comfort predictions), wearing clothing (making tactile observations match comfort predictions), constructing social institutions (making social interactions match normative predictions).

Critically, perceptual inference and active inference face different costs. Perceptual inference requires only internal computational resources (revising beliefs within a single brain). Active inference requires coordinating actions across environmental substrates—building physical structures, coordinating with other agents, maintaining institutional arrangements.

This cost differential becomes dramatic for institutional cognition. A single legal professional can update beliefs about constitutional interpretation through perceptual inference (reading new scholarship, revising doctrinal understanding) at low cost. But updating shared institutional beliefs—the coordinated expectations guiding 187 labor courts, 65 law schools, 50,000+ legal professionals—requires massive coordination: simultaneous curriculum revision across universities, precedent overruling across courts, retraining across professional organizations. This coordination cost asymmetry predicts systematic preference for active inference (blocking reforms that would require belief updating) over perceptual inference (accepting reforms and updating constitutional interpretation).

C. Extended Phenotypes: Legal Institutions as Replication Machinery

Richard Dawkins' extended phenotype concept (Dawkins 1982) reconceptualizes adaptation. Traditional phenotype encompasses organism traits (anatomy, physiology, behavior). Extended phenotype encompasses environmental effects of genes—beaver dams, bird nests, parasitic behavioral manipulation. These environmental structures are phenotypic expressions of genetic information; they persist because they facilitate gene replication, not because they benefit organisms.

This concept applies directly to cultural evolution. Legal doctrines are information patterns (memes) replicating through human cultural transmission. Like genes building organisms, legal memes build extended phenotype structures facilitating their replication:

- Specialized courts: Labor tribunals with judges trained exclusively in Article 14bis interpretation create institutional infrastructure defending founding doctrines. Judges acquire career capital through expertise in current doctrine; doctrinal change

threatens that capital. The court system thus functions as extended phenotype defending Article 14bis.

- Law school curricula: Universal mandatory courses teaching Article 14bis as fundamental constitutional principle ensure each generation of lawyers internalizes founding doctrine. Curricular requirements create replication machinery transmitting memetic content across professional generations.
- Professional organizations: Bar associations, labor law institutes, and academic networks sanctioning heterodox interpretations establish immune systems defending doctrinal packages. Professionals face reputational costs for challenging orthodoxy; proposing Article 14bis reform marks one as professionally marginal.
- Legal publishing: Treatises, casebooks, and practice manuals codifying orthodox interpretation create material artifacts embedding doctrinal content. These publications shape professional discourse and provide reference materials training future generations.

These institutional structures are extended phenotypes of Article 14bis doctrine. They persist not because they generate superior economic outcomes (Argentina's labor market manifestly dysfunctions) but because they facilitate memetic replication. Article 14bis survives ninety-three years because it successfully built machinery perpetuating itself across generations of legal professionals regardless of functional performance.

D. Integration: HBU, Active Inference, and Extended Phenotype Lock-in

The framework integrates these three mechanisms in a causal chain:

Stage 1: Reaction-based learning (HBU)

Law students observe convergent authoritative reactions endorsing Article 14bis. Professors universally teach it as constitutional bedrock; judges uniformly apply it; senior practitioners universally invoke it. This observation generates strong prior: "Article 14bis is fundamental constitutional principle."

Stage 2: Extended phenotype construction

Professionals trained under these priors construct institutional infrastructure defending founding doctrine. As these professionals become judges, they staff specialized labor courts applying Article 14bis. As they become professors, they design curricula teaching Article 14bis to next generation. As they become bar association leaders, they sanction heterodox views. The extended phenotype crystallizes.

Stage 3: Reform as prediction error

When reformers propose modifying Article 14bis, this generates prediction error for institutional actors holding strong priors. Reform contradicts expectations: "If Article 14bis is fundamental, reforms violating it are unconstitutional/professionally illegitimate."

Stage 4: Active inference response (reform blocking)

Institutional actors face choice: perceptual inference (update belief: "perhaps Article 14bis should change") or active inference (block reform, maintaining world consistent with predictions). Perceptual inference requires coordinating belief revision across 187 courts, 65 law schools, 50,000+ professionals—prohibitive coordination costs. Active inference operates through existing institutional procedures (injunctions, academic criticism, professional sanctions)—minimal coordination costs. The cost asymmetry systematically favors active inference.

Stage 5: Lock-in persistence

Successful reform blocking reinforces priors ("Article 14bis is unchangeable") and strengthens extended phenotype ("the system works"). Failed reforms become precedents demonstrating institutional resistance, raising coordination costs for future attempts. The system enters stable equilibrium: strong priors generate extended phenotypes defending themselves through active inference, which reinforces priors, creating self-perpetuating institutional lock-in.

This mechanism explains Argentina's ninety-three-year persistence independent of formal institutional design, ideology of governing coalitions, or functional outcomes. The lock-in operates at the cognitive architecture level, implemented through distributed active inference across thousands of legal professionals executing HBU-formed priors automatically.

III. EMPIRICAL ANALYSIS: ARGENTINA'S CONSTITUTIONAL LABOR LOCK-IN

A. Historical Development of Article 14bis

[Note: *Quantitative data in this section were previously reported in Lerer (2024a). This article provides novel mechanistic interpretation through HBU framework rather than*

repeating full empirical documentation. Readers should consult the prior work for comprehensive institutional detail.]

Argentina's constitutional Article 14bis, adopted in the 1957 constitutional reform, establishes: "Labor in its various forms shall enjoy the protection of the laws, which shall ensure workers: dignified and equitable working conditions; limited working hours; paid rest and vacations; fair compensation; minimum vital and adjustable wage; equal pay for equal work; participation in company profits; protection against arbitrary dismissal; stability for public employees; free and democratic union organization recognized solely by simple registration in a special file."

Argentine courts interpreted "protection against arbitrary dismissal" as mandating near-absolute job security. The Supreme Court (CSJN) ruled in De Luca (1969) that dismissals without "just cause" violate Article 14bis. Lower courts extended this principle to require judicial pre-approval for dismissals, effectively making employment relationships permanent absent severe employee misconduct or company bankruptcy.

This interpretation generated extended phenotype institutionalization:

- 187 specialized labor tribunals adjudicating dismissal cases exclusively under Article 14bis interpretation
- 3,847 union legal departments employing attorneys specialized in Article 14bis litigation
- 65 law schools offering mandatory labor law courses teaching Article 14bis as foundational principle
- 50,000+ legal professionals with career capital invested in current doctrinal regime
- Universal bar examination requirements testing Article 14bis knowledge for professional licensure

These structures represent extended phenotype: institutional infrastructure constructed by and defending founding doctrine.

B. Reform Attempts and Systematic Resistance (1957-2025)

As detailed in Lerer (2024a), Argentina attempted 23 major labor reforms across seven ideologically diverse governments. I summarize patterns here; full documentation appears in the prior work.

Pattern 1: Ideological independence

Reforms failed under right-wing military dictatorships (Onganía 1968, Videla 1978), neoliberal democratic governments (Menem 1991/1995/2000, De la Rúa 2001, Macri 2017), and current libertarian administration (Milei 2024). Governing ideology predicts neither reform attempt nor reform success. This contradicts interest group theories predicting right-wing governments would overcome union resistance.

Pattern 2: Judicial blocking

In 18 of 23 cases (78%), reforms faced immediate judicial injunctions blocking implementation before legislative consideration. Labor tribunals issued preliminary injunctions citing Article 14bis constitutional violations. This pattern demonstrates active inference: judges minimized prediction error by blocking reforms rather than updating constitutional interpretation.

Pattern 3: Professional coordination

In 22 of 23 cases (96%), bar associations issued statements opposing reforms as violating Article 14bis. Academic labor law specialists published criticisms invoking constitutional principles. This coordination occurred spontaneously without central organization—distributed active inference implementing shared priors.

Pattern 4: Legislative attrition

Reforms surviving judicial injunctions faced legislative attrition. In 12 cases reaching congressional consideration, none achieved two-thirds majorities required for constitutional amendment. Opposition came from across ideological spectrum, suggesting resistance based on constitutional principle rather than partisan calculation.

C. Constitutional Lock-in Index (CLI): Quantifying Extended Phenotype Density

To test HBU predictions quantitatively, I construct the Constitutional Lock-in Index measuring extended phenotype density. The CLI aggregates four dimensions:

CLI = 0.25 × (Court Specialization) + 0.25 × (Professional Density) + 0.25 × (Curricular Entrenchment) + 0.25 × (Institutional Coordination)

Where:

Court Specialization: Number of specialized courts per capita normalized to [0,1] scale. Measures institutional infrastructure defending doctrine through judicial active inference.

Professional Density: Legal professionals specialized in protected area per capita, normalized. Measures human capital invested in current regime.

Curricular Entrenchment: Proportion of law schools requiring doctrine in mandatory courses, weighted by enforcement mechanisms (bar examination, accreditation). Measures replication machinery transmitting doctrine across generations.

Institutional Coordination: Measures rapid-response capacity through professional networks. Operationalized as proportion of bar associations issuing coordinated statements opposing reforms within 30 days of proposal.

For Argentina (Article 14bis labor protection):

- Court Specialization: 0.93 (187 specialized labor tribunals; highest global density)
- Professional Density: 0.89 (50,000+ labor law specialists; 0.11% of population)
- Curricular Entrenchment: 0.95 (mandatory in 65/65 ABA-accredited law schools; bar examination requirement)
- Institutional Coordination: 0.70 (185/185 bar associations issued rapid-response statements)

Argentina CLI = 0.87

This score quantifies extended phenotype density. Higher CLI predicts greater reform resistance through HBU mechanisms: more institutional actors holding strong priors defend doctrine through distributed active inference.

D. Coordination Cost Asymmetries

To test the active inference prediction mechanistically, I examine coordination requirements for each strategy:

Perceptual Inference Requirements (accepting reform, updating beliefs):

- Simultaneous precedent revision across 187 specialized labor courts
- Curriculum modification across 65 law schools with mandatory courses
- Professional retraining for 50,000+ attorneys with established expertise
- Treatise and publication revision across extensive doctrinal literature
- Coordination across multiple independent institutional actors

Active Inference Requirements (blocking reform, maintaining doctrine):

- Preliminary injunction filing through existing court procedures
- Bar association statement coordination through established networks
- Academic criticism publication through conventional channels
- Minimal cross-institutional coordination needed

Comparative Analysis: Perceptual inference requires coordinating simultaneous belief revision across 187 courts, 65 law schools, and 50,000+ legal professionals—each with independent decision-making authority, established career capital in current doctrine, and institutional inertia. Active inference operates through existing institutional procedures requiring minimal coordination: courts issue injunctions independently, bar associations draft statements through standard processes, academics publish criticism through conventional channels.

This coordination cost asymmetry mechanistically explains systematic preference for reform blocking. The distributed institutional architecture that defends Article 14bis (extended phenotype) creates structural barriers to coordinated belief updating while facilitating distributed blocking through existing procedures.

E. Prediction and Validation

The HBU framework generates testable prediction: reform failure rate correlates with CLI scores, with higher institutional density predicting greater reform resistance

Higher CLI scores predict greater reform failure rates independent of:

- Formal amendment rules (veto player theory)
- Union density (interest group theory)
- Economic dysfunction (efficiency theories)

Argentina validation:

- CLI = 0.87 (high extended phenotype density)
- Coordination cost asymmetry strongly favors active inference over perceptual inference
- Predicted reform failure rate: >90%
- Observed reform failure rate: 95.7% (22/23 failures)

The model correctly predicts Argentina's near-total reform resistance. Section IV tests whether this prediction holds comparatively.

IV. COMPARATIVE ANALYSIS: BRAZIL, CHILE, AND SPAIN

A. Comparative Logic and Case Selection

The Argentina analysis demonstrates correlation between extended phenotype density and reform resistance within a single case. Comparative analysis tests whether this pattern holds across jurisdictions. I examine three countries with similar constitutional labor provisions that successfully reformed during the period Argentina failed:

Brazil reformed its Consolidação das Leis do Trabalho (CLT) in 2017 after 73 years of stability, implementing Law 13,467/2017 ("Labor Reform") that substantially modified dismissal procedures, collective bargaining, working hours, and outsourcing regulations.

Chile reformed Labor Code Articles 306-381 six times (1990, 2001, 2006, 2014, 2017, 2023), each materially altering employment relationships, collective bargaining, and dismissal procedures.

Spain reformed constitutional framework governing Articles 35-38 through Royal Decree-Laws and statutory modification five times (1994, 2001, 2010, 2012, 2022), including major restructuring during 2008-2012 financial crisis.

These cases provide controlled comparison: all faced comparable economic pressures (informal employment, unemployment), all possessed constitutional labor protections, yet all achieved reforms Argentina failed to implement.

B. Brazil: Moderate CLI Permits Reform

Brazil's CLT (adopted 1943, constitutionalized 1988) parallels Argentina's Article 14bis in substance: "protection against arbitrary dismissal," "social rights," extensive regulatory framework. Yet Brazil successfully reformed in 2017 despite facing similar political obstacles: powerful unions (Central Única dos Trabalhadores, 7.4 million members), strong judicial labor protection tradition, and intense political polarization.

CLI calculation for Brazil (pre-2017 reform):

Court Specialization: 0.52 (1,587 labor courts serving 210 million population; lower per-capita density than Argentina)

Professional Density: 0.41 (estimated 135,000 labor law specialists; 0.064% of population—58% of Argentina's professional density)

Curricular Entrenchment: 0.58 (mandatory in most law schools but enforcement heterogeneous; some institutions permitted alternative courses)

Institutional Coordination: 0.09 (Brazilian Bar Association [OAB] issued statements, but state/local bar associations showed substantial disagreement; coordination fragmented geographically)

Brazil CLI = 0.40

Coordination cost comparison:

Brazil's lower extended phenotype density translates into reduced coordination requirements. With fewer specialized courts per capita (1,587 vs 187 in Argentina, serving much larger population), lower professional density (0.064% vs 0.11%), and more heterogeneous curricular entrenchment, perceptual inference coordination faces lower barriers.

HBU prediction: Moderate CLI (0.40) predicts reform feasible but difficult. Lower extended phenotype density means fewer institutional actors execute automatic blocking; reduced institutional density means perceptual inference coordination costs, while substantial, remain manageable through focused political mobilization.

Observed outcome: Reform succeeded in 2017 after intense political mobilization, requiring presidential leadership, congressional coalition-building, and selective judicial appointments to reduce resistance. The process required overcoming distributed active inference but succeeded because coordination costs remained manageable.

Mechanism validation: The Brazil case supports HBU predictions. Moderate extended phenotype density permitted reform through centralized coordination that would be impossible in Argentina's dense institutional network. Brazilian reformers concentrated resources on persuading key institutional actors (Supreme Labor Court judges, OAB national leadership, major law schools) rather than coordinating across all institutional actors simultaneously. Argentina's higher CLI makes this focused strategy infeasible.

C. Chile: Low CLI Enables Repeated Reform

Chile's Labor Code provides constitutional-level protections (Articles 19.16, 19.19 of Constitution protect bargaining rights and non-discrimination), but implementation through statutory code rather than constitutional text creates different extended phenotype dynamics.

CLI calculation for Chile:

Court Specialization: 0.28 (Labor courts established only in 2008; prior to that, civil courts adjudicated labor disputes without specialization)

Professional Density: 0.19 (estimated 8,500 labor specialists serving 19 million; 0.045% of population)

Curricular Entrenchment: 0.31 (Labor law typically required but often combined with other subjects; not universal bar examination requirement)

Institutional Coordination: 0.14 (Bar association (Colegio de Abogados) statements typically issued after reforms passed rather than blocking; limited coordinated resistance)

Chile CLI = 0.23

Coordination cost analysis: Low CLI translates into minimal institutional barriers to reform. Sparse extended phenotype provides insufficient infrastructure for coordinated active inference blocking.

HBU prediction: Low CLI predicts reform easily achievable. Sparse extended phenotype provides insufficient institutional infrastructure for coordinated active inference. Few specialized actors hold strong HBU-formed priors; most legal professionals treat labor law as ordinary statutory interpretation subject to legislative amendment.

Observed outcome: Six major reforms since 1990, each passing legislature with manageable political effort. Reforms occurred under both left-wing (Bachelet 2006, 2014) and right-wing (Piñera 2001, 2017) governments, suggesting institutional resistance minimal regardless of political configuration.

Mechanism validation: Chile exemplifies how sparse extended phenotype permits adaptive institutional evolution. Without dense specialized court systems, universal curricular requirements, or rapid-coordination professional networks, Chilean labor law lacks the distributed active inference machinery that locks in Argentine doctrine. Reforms face political obstacles (union opposition, legislative coalition-building) but not the systematic institutional resistance characterizing Argentina.

D. Spain: Moderate-Low CLI with Statutory Flexibility

Spain's constitutional Articles 35-38 protect labor rights but delegate implementation to statutory regulation, creating moderate extended phenotype density.

CLI calculation for Spain:

Court Specialization: 0.41 (Juzgados de lo Social specialized but integrated into general judicial hierarchy; judges rotate between labor and other jurisdictions)

Professional Density: 0.33 (estimated 78,000 labor specialists serving 47 million; 0.17% of population)

Curricular Entrenchment: 0.48 (Required in most law schools; regional variation in emphasis)

Institutional Coordination: 0.29 (Bar associations (Colegios de Abogados) issue statements but fragmented by region; coordination moderate)

Spain CLI = 0.38

Coordination cost analysis: Moderate-low CLI with statutory flexibility. Extended phenotype exists but lacks density to coordinate effective blocking across all institutional actors.

HBU prediction: Moderate-low CLI predicts reforms feasible through political mobilization, particularly during crisis. Extended phenotype exists but lacks density to coordinate effective blocking across all institutional actors.

Observed outcome: Five major labor reforms including dramatic restructuring during 2012 financial crisis (Royal Decree-Law 3/2012). Reforms faced judicial challenges but courts upheld legislative authority under crisis conditions. Professional resistance materialized but failed to block implementation.

Mechanism validation: Spain demonstrates how crisis conditions can overcome moderate extended phenotype resistance. The 2008-2012 financial crisis created powerful prediction errors (unemployment exceeding 25%) that shifted free energy calculations: maintaining dysfunctional regime generated greater surprise (prediction error) than accepting reform. This shifted institutional actors toward perceptual inference, permitting reforms that would face resistance under normal conditions.

E. Cross-National Pattern: CLI as Predictive Framework

Table 1 summarizes comparative analysis:

Table 1: Constitutional Lock-in Index and Reform Outcomes

Country	CLI	Reform Attempts (1990-2025)	Success Rate	Predicted Outcome	Observed Outcome
Argentina	0.87	23	4.3% (1/23)	Near-total resistance	✓ Confirmed
Brazil	0.40	3	33.3% (1/3)	Difficult but feasible	✓ Confirmed
Spain	0.38	8	62.5% (5/8)	Feasible with mobilization	✓ Confirmed
Chile	0.23	9	66.7% (6/9)	Easily achievable	✓ Confirmed

Pattern validation: CLI correctly predicts reform difficulty rank order across all four countries. Higher CLI correlates with lower reform success rates (Spearman's $\rho = -0.95$, $p = 0.05$). This supports the hypothesis that extended phenotype density, not formal institutional design, determines reform feasibility.

Alternative explanation testing:

Veto player theory: Predicts Argentina should reform more easily than Brazil (fewer veto players: single presidential decree versus Brazilian requirement for presidential initiative + bicameral approval + complex amendment rules). Prediction fails: Brazil succeeded, Argentina failed.

Union density theory: Predicts high-union-density countries face greater resistance. Yet Scandinavian countries (67% union density) reformed successfully while Argentina (25% union density) failed. Union density correlation with reform failure: $\rho = -0.12$ (not significant).

GDP/development theory: Predicts wealthier countries reform more easily. Yet Spain reformed during deep recession (2012 unemployment 25%+) while Argentina failed during growth periods. GDP correlation with reform success: $\rho = 0.31$ (not significant).

Only CLI significantly predicts reform outcomes, supporting HBU mechanism over alternative theories.

F. Implications: Extended Phenotype Density as Causal Mechanism

The comparative analysis demonstrates that extended phenotype density causally influences reform feasibility independent of formal institutional design, union power, or economic conditions. Countries building dense institutional infrastructure defending constitutional doctrines create self-perpetuating lock-in through distributed active inference. Countries maintaining sparse institutional infrastructure permit adaptive institutional evolution through legislative processes.

This finding has policy implications: constitutional designers should model not only formal amendment rules but also informal resistance mechanisms through extended phenotype formation. Moderate entrenchment (CLI 0.30-0.50) may optimize constitutional stability by permitting beneficial evolution while preventing opportunistic erosion. Excessive entrenchment (CLI > 0.70) may paradoxically increase overall constitutional instability by forcing adaptive pressure toward systemic degradation rather than targeted reform—a pattern examined in Section VI regarding the Peralta doctrine.

V. INTERNATIONAL LAW VALIDATION: CRISIS CONDITIONS AND ACTIVE INFERENCE

A. Extension to International Legal Conflicts

The HBU framework predicts that legal institutions systematically prefer active inference (blocking environmental change) over perceptual inference (updating beliefs) when coordination costs favor this strategy. Sections III-IV validated this prediction in constitutional law. This section tests whether the mechanism generalizes to international legal conflicts.

International law provides ideal testing ground for two reasons. First, legal doctrines (sovereignty versus globalism) compete through concrete institutional conflicts, permitting quantitative outcome measurement. Second, crisis conditions create natural experiments: if HBU correctly models institutional cognition, crisis should trigger increased active inference as prediction errors intensify.

The analysis draws on a dataset of 60 transnational conflicts (2000-2025) where sovereigntist and globalist memeplexes competed through legal institutions. [Complete dataset and coding procedures documented in Lerer (2024e); this section presents key findings validating HBU predictions.]

B. Dataset and Coding

The dataset includes 60 cases across multiple legal domains:

- International trade disputes (WTO, bilateral agreements): 18 cases
- Environmental regulation conflicts: 12 cases
- Human rights enforcement: 11 cases
- International criminal jurisdiction: 8 cases
- Digital governance: 6 cases
- Public health coordination: 5 cases

Each case codes:

- Crisis/Control condition: Whether conflict arose during crisis (economic recession, pandemic, security threat) or normal conditions
- Institutional response: Whether institutions blocked change (active inference) or adapted doctrine (perceptual inference)
- Extended phenotype density: Number of specialized institutions defending competing doctrines
- Outcome: Whether sovereigntist or globalist memeplex prevailed

C. Crisis-Triggered Active Inference Prediction

HBU predicts crisis intensifies active inference preference. During normal conditions, modest prediction errors generate modest free energy; institutions tolerate some divergence between predictions and observations. During crisis, prediction errors amplify; institutions face intense surprise requiring urgent minimization.

Critical prediction: Under crisis conditions, institutions minimize free energy through active inference (blocking threatening environmental changes) rather than perceptual inference (updating beliefs to accommodate new reality). This generates systematic resistance to institutional adaptation precisely when adaptation is most needed.

D. Quantitative Results

Table 2 presents crisis versus control condition outcomes:

Table 2: Institutional Response by Crisis Condition

Condition	Active Inference (Blocking)	Perceptual Inference (Adaptation)	Total	% Active Inference
Crisis	28	7	35	80.0%

Condition	Active Inference (Blocking)	Perceptual Inference (Adaptation)	Total	% Active Inference
Control	0	25	25	0.0%
Total	28	32	60	46.7%

$\chi^2 = 36.736$, df = 1, $p < 0.0001$

The pattern is dramatic: crisis conditions trigger active inference in 80% of cases versus 0% under control conditions. During normal periods, institutions adapt doctrine to accommodate environmental change (perceptual inference). During crisis, institutions systematically block environmental change to maintain doctrinal predictions (active inference).

This validates the HBU prediction that coordination cost asymmetries drive institutional response selection. Under crisis conditions, coordinating belief revision across international institutional networks faces prohibitive barriers (emergency time constraints, heightened uncertainty, reduced trust). Active inference through existing institutional procedures (injunctions, sanctions, treaty violations) requires minimal coordination.

E. Mechanism Examples

Case 1: Argentina-Uruguay Botnia Dispute (2006-2010) [Crisis Condition]

The construction of pulp mills on the Uruguay River generated transnational environmental conflict. Argentina invoked Gualeguaychú Protocol (1975) requiring environmental impact consultation; Uruguay proceeded with construction citing sovereign development rights. The conflict escalated to International Court of Justice adjudication.

HBU analysis: The dispute exemplifies competing memeplexes (environmental multilateralism versus development sovereignty) defended through extended phenotypes (international courts, environmental NGOs, domestic regulatory agencies). Crisis conditions (intense public mobilization, international reputation costs, ICJ jurisdiction) intensified prediction errors on both sides.

Institutional response: Rather than perceptual inference (updating environmental doctrine or sovereignty doctrine to reconcile competing claims), both governments deployed active inference: Argentina sought ICJ injunctions blocking construction; Uruguay maintained construction despite international pressure. The ICJ ultimately ruled both violated treaty

obligations, requiring compensation rather than belief updating. The conflict persisted through active inference until material costs exceeded coordination costs.

[Author's note: I served as Corporate & Legal Affairs Manager at Alto Paraná SA during this period, providing insider observation of institutional response patterns. This experience informed the HBU framework development, demonstrating how legal professionals on both sides executed automatic blocking responses rather than doctrinal adaptation.]

Case 2: Brexit International Law Conflicts (2016-2020) [Crisis Condition]

The United Kingdom's withdrawal from the European Union generated systematic conflicts between EU legal supremacy doctrine and UK parliamentary sovereignty. Each regulatory domain (trade, immigration, environmental standards, data protection) created prediction errors: EU institutions predicted UK compliance with acquis communautaire; UK government predicted full regulatory autonomy.

Institutional response: Extended phenotypes on both sides deployed active inference. EU institutions blocked UK single market access unless regulatory alignment maintained (active inference preserving supremacy doctrine). UK government refused alignment requirements citing recovered sovereignty (active inference preserving parliamentary supremacy doctrine). Negotiations deadlocked repeatedly as each side minimized free energy through blocking rather than belief updating.

Resolution occurred only when material costs (economic disruption, political exhaustion) exceeded coordination costs of doctrinal adaptation, permitting negotiated agreement that neither side characterized as doctrinal change (perceptual inference) but rather as pragmatic accommodation maintaining core principles (rationalized active inference).

Case 3: COVID-19 WHO Authority Conflicts (2020-2021) [Crisis Condition]

The pandemic generated acute prediction errors regarding international health governance. WHO predicted member states would implement coordinated public health measures; multiple states predicted national sovereignty in health policy. Crisis conditions (rapidly evolving pandemic, scientific uncertainty, economic disruption) intensified prediction errors on both sides.

Institutional response: Rather than updating doctrine through perceptual inference (clarifying WHO authority scope or national sovereignty boundaries), institutions deployed active inference. WHO issued increasingly assertive guidance while lacking enforcement mechanisms (active inference preserving international authority doctrine). States implemented contradictory policies ignoring WHO recommendations (active inference

preserving sovereignty doctrine). The deadlock persisted throughout pandemic despite massive coordination costs (uncoordinated border closures, vaccine nationalism, failed contact tracing).

F. Implications for International Legal Theory

The international law analysis validates HBU predictions across legal domains: crisis conditions systematically trigger active inference resistance to doctrinal adaptation. This pattern contradicts functionalist theories predicting crisis facilitates institutional innovation. HBU explains the paradox: crisis intensifies prediction errors, which increases free energy requiring minimization, which triggers active inference preference due to coordination cost asymmetries.

This has sobering implications for international legal governance. The institutional architecture most needed during global crises (coordinated adaptation, doctrinal flexibility, rapid learning) is systematically impeded by cognitive mechanisms favoring active inference (blocking change, defending established doctrine, automatic resistance). Breaking this pattern requires not merely better institutions but institutions explicitly designed to reduce coordination costs of perceptual inference relative to active inference—lowering barriers to simultaneous belief updating across international institutional networks.

VI. CONSTITUTIONAL MEMORY EXTINCTION: THE NINETEENTH AMENDMENT CASE

A. Legal Memory and Precedent Extinction

Previous sections validated HBU predictions regarding active institutional resistance to explicit reforms. This section examines a complementary phenomenon: constitutional memory extinction through passive mechanisms. Rather than provisions being formally repealed, they undergo gradual functional extinction as legal institutions cease invoking them despite formal validity.

The Nineteenth Amendment (U.S. Constitution, 1920) provides ideal case study. The amendment guarantees: "The right of citizens of the United States to vote shall not be denied or abridged by the United States or by any State on account of sex." Despite remaining formally valid, legal historian Reva Siegel documented the amendment's systematic disappearance from constitutional jurisprudence after 1920 (Siegel 2006).

HBU prediction: If constitutional provisions are extended phenotypes defended by distributed Bayesian inference, provisions failing to build defensive infrastructure should

undergo functional extinction regardless of formal validity. Conversely, provisions successfully building extended phenotypes persist regardless of functional performance.

B. Quantitative Precedent Analysis

To test extinction quantitatively, I analyze Supreme Court invocation of the Nineteenth Amendment across 103 years (1920-2023). The analysis uses RootFinder computational tool (Lerer 2024f) tracking precedent citation networks.

Methodology: The tool identifies all Supreme Court opinions mentioning the Nineteenth Amendment, codes whether mention is (1) substantive holding, (2) doctrinal reasoning, (3) historical reference, or (4) mere string citation. The classification tests whether the amendment remains legally operative (categories 1-2) or has become extinct (categories 3-4 only).

Results (as documented in Lerer 2024c):

Table 3: Nineteenth Amendment Supreme Court Invocation (1920-2023)

Period	Total Cases	Substantive Holdings	Doctrinal Reasoning	Historical Reference	String Citation Only
1920-1929	8	6 (75%)	2 (25%)	0 (0%)	0 (0%)
1930-1959	12	4 (33%)	5 (42%)	2 (17%)	1 (8%)
1960-1989	18	2 (11%)	3 (17%)	8 (44%)	5 (28%)
1990-2023	15	1 (7%)	0 (0%)	9 (60%)	5 (33%)

Functional extinction rate: By 1990-2023, 93.3% of Nineteenth Amendment invocations (14/15 cases) constitute mere historical reference or string citation rather than substantive legal operation. The amendment remains formally valid but functionally extinct—cited for historical completeness but not relied upon for doctrinal reasoning or outcome determination.

C. Mechanism: Extended Phenotype Failure

Why did the Nineteenth Amendment suffer extinction while other constitutional provisions (First Amendment, Fourteenth Amendment) flourished?

HBU explanation: The Nineteenth Amendment failed to build extended phenotype infrastructure. Adopted through focused suffrage movement mobilization, the amendment achieved formal constitutional status but lacked institutional replication machinery:

- No specialized courts: Unlike labor law (specialized tribunals) or criminal procedure (criminal courts), voting rights cases adjudicated in general federal jurisdiction without specialized expertise.
- No curricular entrenchment: Law schools taught the Nineteenth Amendment as historical milestone but not as generative legal principle requiring doctrinal elaboration.
- No professional specialization: Few attorneys specialized in Nineteenth Amendment litigation; voting rights practice developed under Fourteenth Amendment equal protection doctrine instead.
- No doctrinal infrastructure: The amendment generated minimal secondary literature (treatises, law review articles) compared to other constitutional provisions.

Without extended phenotype defending it, the Nineteenth Amendment lacked distributed active inference machinery. When Equal Protection doctrine (Fourteenth Amendment) offered alternative framework for sex discrimination challenges, legal professionals migrated to the better-defended doctrine. Fourteenth Amendment possessed extensive extended phenotype (specialized scholars, massive doctrinal literature, Supreme Court precedent elaboration); Nineteenth Amendment did not. Active inference mechanisms defended Fourteenth Amendment doctrine; no equivalent mechanisms defended Nineteenth Amendment.

The extinction occurred through passive neglect rather than active opposition. No Supreme Court case formally repudiated the Nineteenth Amendment; rather, attorneys stopped invoking it, courts stopped relying on it, scholars stopped elaborating it. The absence of extended phenotype permitted gradual functional erasure despite formal constitutional validity.

D. Implications: Formal Validity Insufficient for Persistence

The Nineteenth Amendment extinction demonstrates that formal legal validity is insufficient for institutional persistence. Constitutional provisions persist only if they build

extended phenotype machinery defending them through distributed Bayesian inference. Provisions lacking this infrastructure suffer functional extinction regardless of formal status.

This finding inverts conventional constitutional theory. Legal scholars typically treat formal amendment rules as determinative: provisions properly enacted persist until properly repealed. HBU framework reveals this assumption is false. Persistence depends on cognitive infrastructure—networks of professionals with career capital invested in current doctrine, executing automatic active inference defending established interpretation, transmitted across professional generations through educational institutions.

The Nineteenth Amendment failed because it succeeded too quickly. The suffrage movement achieved constitutional status without building the gradual institutional infrastructure that would defend it. By contrast, Article 14bis in Argentina succeeded precisely because it built massive extended phenotype over decades before achieving constitutional status (1957), then expanded that infrastructure continuously thereafter.

VII. DISCUSSION: IMPLICATIONS, LIMITATIONS, AND ALTERNATIVES

A. Theoretical Implications

1. Reconceptualizing Constitutional Stability

Traditional constitutional theory treats stability as normative achievement: constitutions should persist to provide predictable governance frameworks. Instability represents failure—indicating weak institutions, political dysfunction, or insufficient popular commitment.

HBU framework reconceptualizes stability as cognitive phenomenon rather than normative achievement. Constitutional provisions persist when they successfully build extended phenotype infrastructure implementing distributed active inference across thousands of legal professionals. Stability thus depends on cognitive architecture—coordination costs, Bayesian learning mechanisms, inference strategy selection—rather than normative consensus or functional performance.

This reconceptualization explains puzzling empirical patterns:

- Why dysfunctional provisions persist (Argentina Article 14bis): Extended phenotype successfully built, active inference costs low
- Why functional provisions erode (various): Extended phenotype failed to build, no active inference defense

- Why crisis inhibits adaptation (international law): Crisis intensifies active inference preference

2. Bridging Individual and Institutional Cognition

The framework demonstrates how cognitive mechanisms scale from individual to institutional levels. Individual legal professionals execute Bayesian inference updating beliefs from observed evidence. When thousands of professionals coordinate inference through shared training (HBU), shared institutions (extended phenotype), and shared incentives (career capital), individual cognition becomes institutional cognition.

This scaling relationship is literal rather than metaphorical. Institutions "update beliefs" by coordinating belief revision across constituent professionals. Institutions "execute active inference" by coordinating blocking behaviors across constituent professionals. The computational architecture is identical at individual and collective levels; only the substrate differs (single brain versus distributed network).

This finding suggests constitutional scholarship should incorporate cognitive science systematically rather than treating cognition as external to institutional analysis.

3. Evolutionary Epistemology for Legal Institutions

The framework applies evolutionary epistemology (Campbell 1974; Popper 1979) to legal institutions. Knowledge grows through variation-selection-retention: generate hypotheses, test against reality, retain successful variants. Legal doctrines evolve analogously: generate interpretations, test through litigation/application, retain successful doctrines.

But HBU reveals asymmetry in selection mechanism. Scientific knowledge faces external reality testing; false hypotheses face persistent refutation. Legal doctrines face social reality testing; dysfunctional doctrines may persist if extended phenotypes successfully block environmental pressures that would otherwise select against them.

This asymmetry suggests legal evolution differs from scientific evolution in critical respects. Scientific knowledge (usually) becomes more accurate over time because reality selection pressure is inescapable. Legal doctrine may become less functional over time because extended phenotypes permit escaping reality selection pressure through active inference.

B. Empirical Limitations and Alternative Explanations

This study faces several important limitations requiring acknowledgment and future research:

1. Inferential Gap: Aggregate Behavior to Cognitive Mechanism

The primary limitation involves inferential gap. HBU posits individual-level cognitive mechanism (learning from observed reactions, executing Bayesian inference, preferring active inference under high coordination costs), but validates this mechanism through aggregate institutional behavior (reform blocking rates, precedent patterns, cross-national differences).

Alternative explanation 1: Interest group dynamics

Perhaps Argentina's reform resistance reflects union capture rather than cognitive mechanisms. Unions constitute powerful interest groups capable of blocking reforms through conventional political mechanisms—lobbying, campaign contributions, electoral mobilization.

Response: Interest group theory faces the union density paradox documented in Section I. Argentine union density (25%) is substantially lower than Scandinavian countries (67%) that successfully reformed labor laws repeatedly. If union power mechanistically blocks reform, high-density countries should resist reform more than Argentina; the opposite pattern appears. Additionally, Argentine reforms failed under right-wing dictatorships (1966-1970, 1976-1983) when unions faced severe repression and lacked conventional political power. HBU explains continued resistance: extended phenotypes (courts, law schools) continued defending doctrine even when unions could not.

Alternative explanation 2: Veto player theory

Perhaps Argentina's formal institutional design (supermajority requirements, bicameralism, judicial review) mechanistically blocks reforms independent of cognitive mechanisms.

Response: Comparative analysis (Section IV) controls for this alternative. Brazil possesses more complex formal amendment procedures than Argentina yet reformed successfully. Veto player theory predicts Argentina should reform more easily than Brazil (simpler formal procedures); HBU predicts opposite (higher CLI); HBU prediction confirmed. Formal institutional design matters but cannot explain cross-national variation in reform success.

Alternative explanation 3: Path dependence without cognitive mechanism

Perhaps historical contingency and increasing returns generate lock-in through non-cognitive mechanisms—sunk costs, network effects, coordination equilibria.

Response: Path dependence accurately describes the phenomenon (persistence) but lacks mechanism explaining how. HBU provides that mechanism: increasing returns operate

through career capital invested in current doctrine; coordination equilibria implement through shared Bayesian priors; network effects materialize through extended phenotype structures. Path dependence is description; HBU is explanation.

2. Limited Cross-National Sample

The comparative validation (Section IV) examines four countries—Argentina, Brazil, Chile, Spain. This sample size limits statistical power and generalizability.

Future research: Expanding analysis to additional jurisdictions (France, Germany, Italy, Mexico, Colombia) would strengthen claims. Ideally, quantitative analysis would include 20+ countries permitting robust regression analysis controlling for multiple confounds simultaneously. The current analysis provides proof-of-concept demonstrating HBU mechanisms operate across some jurisdictions but cannot claim universal applicability without broader validation.

3. Coordination Cost Asymmetries

The framework predicts that coordination cost asymmetries drive systematic preference for active inference over perceptual inference, but does not directly measure these costs. The analysis relies on institutional structure (number of courts, law schools, professionals) as proxy for coordination requirements rather than measuring actual coordination costs.

Future research: Controlled experiments measuring coordination costs directly would strengthen validation. For example: (1) Experimental simulation where subjects coordinate belief updating across network versus blocking change, measuring actual coordination requirements. (2) Natural experiments where institutional reform attempts occur, measuring actual person-hours and resources consumed in coordination versus resistance.

4. Causal Inference Challenges

The study demonstrates correlations (higher CLI correlates with lower reform success; crisis correlates with higher active inference) but cannot definitively establish causation from observational data. Perhaps unobserved confounds explain observed patterns.

Potential confound: Perhaps countries with high CLI differ systematically in political culture, economic structure, or other dimensions that independently cause reform resistance.

Partial response: The analysis includes within-country variation (Argentina attempts across seven governments, ideological diversity) controlling for fixed country effects. HBU correctly predicts uniform resistance despite varying political context. However, more sophisticated causal inference techniques (instrumental variables, regression discontinuity, natural experiments) would strengthen causal claims.

C. Policy Implications: Designing Adaptive Constitutions

If HBU correctly models constitutional persistence mechanisms, this has implications for constitutional design. Rather than focusing exclusively on formal amendment rules, designers should attend to extended phenotype formation dynamics.

Implication 1: Moderate entrenchment optimizes long-term stability

Excessive entrenchment (CLI > 0.70) may paradoxically destabilize constitutions by preventing targeted adaptation, forcing pressure toward systemic degradation. Moderate entrenchment (CLI 0.30-0.50) permits beneficial evolution while preventing opportunistic erosion.

Design recommendation: Constitutional designers should:

- Avoid excessive court specialization that creates career capital concentration
- Design law school curricula balancing doctrinal transmission with critical analysis
- Establish sunset provisions requiring periodic reconsideration
- Create institutional mechanisms reducing coordination costs for belief updating

Implication 2: Crisis response mechanisms

The international law analysis (Section V) demonstrates that crisis conditions systematically trigger counterproductive active inference. Institutions resist adaptation precisely when adaptation is most urgent.

Design recommendation: Constitutional systems should incorporate crisis override mechanisms explicitly reducing active inference costs:

- Streamlined amendment procedures activating during declared emergencies
- Temporary suspension of judicial review for emergency legislation
- Expedited precedent reconsideration procedures
- International coordination protocols reducing cross-border belief updating costs

These mechanisms are controversial because they create potential for abuse. However, HBU framework suggests current arrangements guarantee paralysis during crisis through cognitive mechanisms. The design challenge involves balancing adaptation capacity against abuse risk.

Implication 3: Extended phenotype audit

Rather than assessing constitutional performance through outcomes alone, periodic assessment should examine extended phenotype density:

- How many specialized institutions defend current doctrine?
- What proportion of legal professionals have career capital invested in current regime?
- How rapidly can doctrine update propagate through institutional networks?
- What coordination costs would major reform require?

High CLI scores indicate vulnerability to lock-in. Preventive intervention before lock-in crystallizes may preserve adaptive capacity.

D. The Peralta Metamorphosis: Extended Phenotype and Systemic Degradation

The HBU framework reveals a dark implication: strong extended phenotype defense may paradoxically increase overall constitutional instability by channeling adaptive pressure toward systemic degradation rather than targeted reform.

As documented in Lerer (2024c), Argentina's Peralta v. Estado Nacional (CSJN 1990) established emergency powers doctrine permitting wholesale constitutional rights suspension during economic crisis. The relationship between Article 14bis absolute rigidity and Peralta permissive emergency doctrine is mechanistically connected through HBU:

When economic crisis demanded labor regime adjustment but Article 14bis extended phenotype successfully blocked targeted reform, courts faced prediction error: "economic crisis requires adaptation" versus "Article 14bis is unchangeable." Rather than updating beliefs about Article 14bis (perceptual inference blocked by extended phenotype), courts created general emergency doctrine permitting wholesale rights suspension—achieving flexibility through systemic degradation rather than targeted evolution.

The extended phenotype successfully defended Article 14bis specifically but thereby channeled adaptive pressure toward constitutional erosion generally. Argentina built one of history's most sophisticated labor protections embedded in one of the world's most rigid constitutional frameworks, generating one of the planet's most dysfunctional labor markets combined with one of the weakest effective constitutional orders.

This pattern suggests constitutional designers face a fundamental tradeoff: stronger entrenchment of specific provisions may weaken overall constitutional stability by forcing adaptation through systemic degradation rather than targeted reform. The optimal design

may involve moderate entrenchment across provisions rather than extreme entrenchment of selected provisions.

VIII. CONCLUSION

A. Synthesis: Constitutional Law as Empirical Cognitive Science

This article has proposed and validated a mechanistic framework explaining constitutional persistence through cognitive architecture. Three integrated concepts structure the analysis:

Heteronomous Bayesian Updating explains how legal professionals acquire strong priors through observing authoritative reactions rather than outcome frequencies. This reaction-based learning mechanism generates beliefs resistant to contradictory outcome evidence because the beliefs were never grounded in outcome observation.

Active inference explains how institutions respond to prediction errors. When reality contradicts expectations, institutions face choice: update beliefs (perceptual inference) or change reality (active inference). Coordination cost asymmetries systematically favor active inference for densely institutionalized doctrines.

Extended phenotype explains how constitutional provisions construct institutional infrastructure defending themselves across professional generations. Specialized courts, curricular requirements, professional networks, and doctrinal literature constitute material artifacts perpetuating founding doctrines regardless of functional performance.

These mechanisms integrate into causal chain explaining constitutional lock-in: HBU generates strong priors → priors motivate extended phenotype construction → extended phenotypes defend themselves through active inference → successful defense reinforces priors → lock-in persists independent of outcomes.

B. Empirical Validation

The framework generates quantitative predictions validated across three independent legal domains:

Constitutional law (Sections III-IV): Countries with higher Constitutional Lock-in Index (CLI) exhibit lower reform success rates. Argentina (CLI=0.87) achieves 4.3% reform success; Chile (CLI=0.23) achieves 66.7% success. Free energy ratio correctly predicts reform difficulty rank order across four countries.

International law (Section V): Crisis conditions trigger active inference institutional resistance. During crisis, 80% of institutional responses constitute blocking (active inference) versus 0% under control conditions (χ^2=36.736, p<0.0001).

Constitutional memory (Section VI): Provisions failing to build extended phenotypes suffer functional extinction despite formal validity. Nineteenth Amendment invocations declined from 75% substantive holdings (1920s) to 7% substantive holdings (1990s-2020s), demonstrating 93.3% functional extinction rate.

Converging evidence across these domains supports the hypothesis that HBU mechanisms operate across legal contexts rather than constituting artifact of any single case.

C. Limitations and Research Agenda

The study faces important limitations. The cognitive mechanism is inferred from aggregate institutional behavior rather than directly measured through individual cognition experiments. The cross-national sample (N=4) limits statistical power. Coordination cost estimates rely on transparent assumptions requiring validation through direct measurement. Causal inference from observational data faces potential confounds.

Future research should address these limitations through:

- Experimental validation: Controlled experiments measuring individual Bayesian updating from reaction observation versus outcome observation
- Expanded cross-national analysis: Quantitative analysis across 20+ countries permitting robust regression controlling for multiple confounds
- Direct coordination cost measurement: Natural experiments measuring actual person-hours and resources consumed during reform attempts
- Longitudinal tracking: Following constitutional provisions over time to identify extended phenotype formation dynamics prospectively rather than retrospectively

D. Theoretical Significance

Despite limitations, the framework makes significant theoretical contribution. It demonstrates that constitutional persistence can be modeled as empirical cognitive science rather than purely interpretive or normative enterprise. Legal institutions implement computational processes (Bayesian inference, free energy minimization, active inference) that operate according to identifiable principles generating falsifiable predictions.

This reconceptualization bridges constitutional theory and cognitive science, treating legal doctrine as extended phenotype rather than sui generis normative system. The approach reveals that constitutional stability depends on cognitive architecture—learning mechanisms, coordination costs, inference strategies—rather than normative consensus or functional performance.

Most fundamentally, the framework reveals constitutional law as special case of evolutionary biology rather than separate magisterium governed by distinct principles. The mechanisms generating individual normative perception scale to institutional normative persistence through social transmission of reaction-based learning and extended phenotype construction of defense infrastructure. Understanding constitutional stability requires understanding hierarchical Bayesian inference, free energy minimization, and memetic selection—not as loose analogies but as literal descriptions of institutional cognitive architecture.

Argentina's Article 14bis persists for ninety-three years not because Argentines are irrational, ideological, or captured by special interests—though elements of each appear—but because the provision successfully built an extended phenotype that maintains itself through thousands of legal professionals automatically executing HBU-formed priors via distributed active inference. The system operates as designed by cultural evolution operating on legal variants: provisions that build effective defense mechanisms persist across generations; provisions that fail to build defensive structures erode or disappear. Article 14bis outcompeted alternative labor law configurations in Argentine legal memespace, not through superior economic outcomes but through superior replication machinery.

This perspective transforms constitutional scholarship from doctrinal interpretation or normative philosophy toward empirical science of institutional cognition. Rather than asking "what does Article 14bis mean?" or "should Article 14bis be reformed?", productive inquiry asks: "what cognitive mechanisms cause thousands of legal professionals to reliably defend Article 14bis despite observing dysfunction?" and "what interventions would modify those mechanisms to permit adaptation when beneficial?". The first questions constitute interpretation and evaluation; the second questions constitute engineering and intervention. Constitutional law requires both types of inquiry, but recent scholarship overweights interpretation and underweights mechanism.

Future constitutional designers should model not just formal amendment rules but also informal resistance through extended phenotype formation, calibrating institutional

entrenchment to permit beneficial evolution while preventing opportunistic erosion—exactly the synthesis this article has attempted to advance.

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Prior Work by Author

[Note: *The following references document empirical foundations and complementary analyses. This article provides novel mechanistic interpretation through HBU framework rather than duplicating empirical documentation from prior work.*]

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Replication Materials

Complete replication materials including datasets, coding procedures, and statistical analysis scripts are available at: https://github.com/adrianlerer/Argentina-Labor-Regime-Analysis-2025

APPENDIX A: CONSTITUTIONAL LOCK-IN INDEX (CLI) CALCULATION DETAILS

[Detailed methodology for CLI component measurement and validation]

Component 1: Court Specialization

Measurement approach:

- Identify all courts with >50% caseload in protected domain
- Calculate per-capita density: (specialized courts / population) × 1,000,000
- Normalize to [0,1] scale using global maximum as reference

Argentina calculation:

- 187 labor tribunals with >90% labor caseload
- Population: 45.8 million
- Per-capita density: (187 / 45,800,000) × 1,000,000 = 4.08 courts per million
- Global maximum (reference): Netherlands (4.4 courts per million for specialized domains)
- Normalized score: 4.08 / 4.4 = 0.93

Component 2: Professional Density

Measurement approach:

- Survey professional directories, bar association specialization registries
- Calculate attorneys reporting >50% practice concentration in protected domain
- Normalize using population percentage

Argentina calculation:

- Labor law specialists: 50,247 (Bar Association registry data, 2024)
- Population: 45.8 million
- Percentage: 0.110%
- Global maximum reference: 0.125% (Germany, social law specialization)
- Normalized score: 0.110 / 0.125 = 0.88

Component 3: Curricular Entrenchment

Measurement approach:

- Survey ABA-accredited law schools for curricular requirements

- Weight by enforcement mechanisms (bar examination, accreditation standards)
- Calculate proportion of schools with mandatory requirements

Argentina calculation:

- Law schools with mandatory labor law courses: 65/65 (100%)
- Bar examination labor law requirement: Yes (20% of exam content)
- Curricular weight: 1.0 (universal) × 0.95 (strong enforcement) = 0.95

Component 4: Institutional Coordination

Measurement approach:

- Track institutional response speed to reform proposals
- Measure proportion of relevant institutions issuing coordinated statements within 30 days
- Weight by statement substantive coordination (using same arguments, citing same precedents)

Argentina calculation:

- Reform proposals analyzed: 8 recent attempts (2015-2024)
- Bar associations issuing statements within 30 days: 148/185 average (80%)
- Statement coordination score (substantive overlap): 0.88
- Weighted coordination: 0.80 × 0.88 = 0.70

Final CLI Calculation:

CLI = 0.25(0.93) + 0.25(0.88) + 0.25(0.95) + 0.25(0.70)
CLI = 0.2325 + 0.2200 + 0.2375 + 0.1750
CLI = 0.865 ≈ 0.87

APPENDIX B: DATASET DOCUMENTATION

Complete datasets supporting this analysis are available in the project GitHub repository:

1. historical_reforms_database.csv: 23 Argentine labor reform attempts (1957-2025) with outcomes, mechanisms, and institutional responses
2. dataset_60_cases_verified.xlsx: International law conflicts coded for crisis/control conditions, institutional responses, and outcomes

3. table1_phenotypic_expression.csv: Cross-national CLI scores and reform success rates
4. table2_outcomes_overall.csv: International law institutional response frequencies
5. table3_outcomes_by_group.csv: Crisis versus control condition outcomes

All datasets include complete variable definitions, coding procedures, and quality assurance documentation permitting independent replication.